

Exercise 04: Firewalls and IDS

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We learned Firewall and IDS today. I am bad at remembering things like names and concepts, for example: *cafe latte*, *cappuccino*, *cortado*, and *macchiato*. What about you?

1 Make an Easy Explanation to Your Friend Who wasn't Here Today (15 min)

It is hard to remember many things. Therefore, I always think of concrete examples or metaphors. Please think about a nice metaphor for each concept you learned today.

Firewalls

Try to explain the different types of firewall that we have learned today. Packet Filtering Firewall, Stateful Firewall, Application-Level Gateway/Proxy Firewall. Also, find other types of firewalls we did not touch today.

2 Brute Force Attack on a Hashed Password (Self-study: 40-80 min)

It is also hard to remember a long password, but wait! If you use a short, frequently used, popular password, it is dangerous.

Objective

Your task is to crack this password that has been hashed using the *BCRYPT algorithm*. Try to find the password using the *rockyou.txt* wordlist.¹ Ironically, this wordlist was compiled after a breach of the social app RockYou in 2009, where unencrypted user passwords were stolen.

You may use any tool you prefer, such as **Hashcat** or **John the Ripper**, or write your own script to complete the task.

Hashed password

\$2b\$12\$SoreoddaNP22fhRp00DWpeOD8USnAHjzFjRfKh1QyVzmTxikhL0lm

Requirements

- Use the **rockyou.txt** wordlist to crack the hashed password.
- Choose any tool or method you find suitable.

¹pre-included in Kali Linux